Diabetes Mellitus

All key points for the exam (ADA Standards of Care) — drug tables included

1. Classification and Pathophysiology

Type	Key features
Type 1 (T1DM) (~5–10%)	**Autoimmune β-cell destruction → absolute insulin deficiency** • usually children/young adults (any age) • lean • **prone to DKA** • autoantibodies: **GAD65**, **IA-2**, **ZnT8**, insulin autoantibodies • **low C-peptide** • needs insulin from diagnosis
Type 2 (T2DM) (~90–95%)	**Insulin resistance** + **progressive β-cell dysfunction** (relative deficiency) • adults (increasingly youth) • **obesity**, family history, inactivity • gradual onset, often asymptomatic • HHS more than DKA
Gestational (GDM)	Diabetes first diagnosed in the **2nd or 3rd trimester** that was not clearly overt before pregnancy
Specific types	**MODY** (monogenic, autosomal dominant, young, non-obese — HNF1A/HNF4A respond to **sulfonylureas**; GCK → no treatment), **LADA** (slow autoimmune in adults), pancreatic disease (cystic fibrosis, pancreatitis, pancreatectomy), endocrinopathies (Cushing's, acromegaly), **drug-induced** (glucocorticoids, antipsychotics, post-transplant)

Stages of type 1 diabetes

- **Stage 1:** ≥ 2 islet autoantibodies + normoglycemia (presymptomatic).
- **Stage 2:** autoantibodies + **dysglycemia** (presymptomatic) → **teplizumab** can delay stage 3 (age ≥ 8).
- **Stage 3:** clinical (symptomatic) diabetes.

Pathophysiology of T2DM ("ominous octet")

- ↓ insulin secretion (β-cell) • ↑ glucagon (α-cell) • ↑ hepatic glucose production • ↓ muscle glucose uptake • ↑ lipolysis (adipose) • ↓ incretin effect (GLP-1/GIP) • ↑ renal glucose reabsorption (SGLT2) • neurotransmitter dysfunction (appetite).
- Each drug class targets one or more of these defects.

Symptoms

- **Polyuria, polydipsia, polyphagia**, unexplained **weight loss** (T1), fatigue, **blurred vision**, slow wound healing, recurrent infections (candidiasis, UTI).

2. Screening and Diagnosis

Test	Normal	Prediabetes	Diabetes
A1C	< 5.7%	**5.7–6.4%**	**≥ 6.5%**
Fasting plasma glucose (≥ 8 h no calories)	< 100 mg/dL	**100–125** (IFG)	**≥ 126 mg/dL**
2-h plasma glucose (75-g OGTT)	< 140 mg/dL	**140–199** (IGT)	**≥ 200 mg/dL**
Random plasma glucose	—	—	**≥ 200** + classic symptoms or hyperglycemic crisis

- Without unequivocal hyperglycemia, diagnosis needs **2 abnormal results** (same sample or 2 separate samples).
- **A1C is unreliable** with: anemia (iron deficiency ↑, hemolysis ↓), hemoglobinopathies (sickle cell), **pregnancy** (2nd/3rd trimester), recent blood loss or **transfusion**, **EPO therapy**, **ESRD/hemodialysis**, HIV, G6PD deficiency → use glucose-based criteria.
- **A1C** reflects average glucose over **~2–3 months** • **eAG = 28.7 × A1C − 46.7** (A1C 6% = 126, **7% = 154**, 8% = 183, 9% = 212 mg/dL).
- Each 1% change in A1C ≈ 28–30 mg/dL change in average glucose.

Who to screen (ADA)

- **All adults ≥ 35 years** • any age if **overweight/obese (BMI ≥ 25, or ≥ 23 in Asian Americans)** + ≥ 1 risk factor: first-degree relative, high-risk ethnicity, **CVD**, **HTN** (≥ 130/80 or on therapy), HDL < 35 or TG > 250, **PCOS**, physical inactivity, acanthosis nigricans, HIV.
- **Prediabetes → test yearly • history of GDM → at least every 3 years lifelong** • normal → repeat every **3 years**.
- People on drugs that raise glucose (**glucocorticoids**, **atypical antipsychotics**, statins, thiazides, HIV drugs) and after pancreatitis → screen.
- **Youth:** screen after puberty or ≥ 10 years if overweight + risk factors.
- T1DM: autoantibody screening for relatives of T1DM patients (research/TrialNet).

3. Prediabetes and Prevention of T2DM

- **Intensive lifestyle (Diabetes Prevention Program): ≥ 7% weight loss** + **≥ 150 min/week** moderate activity → **↓ T2DM by 58%** (more effective than metformin, 31%).
- **Metformin** for prevention: especially **BMI ≥ 35**, **age < 60**, and women with **prior GDM** • monitor vitamin B12.
- GLP-1 RAs/tirzepatide (weight loss) and pioglitazone also reduce progression • treat CV risk factors • annual monitoring.

4. Glycemic Goals and Monitoring

Parameter	Goal (most nonpregnant adults)
A1C	**< 7%** • **< 6.5%** if achievable without hypoglycemia (young, short duration, no CVD) • **< 8%** (less stringent) for limited life expectancy, severe hypoglycemia history, advanced complications, many comorbidities
Preprandial (fasting) glucose	**80–130 mg/dL**
Peak postprandial (1–2 h after start of meal)	**< 180 mg/dL**
CGM — Time in range (70–180 mg/dL)	**> 70%** (≈ A1C 7%) • older/high risk > 50%
CGM — Time below range	**< 70 mg/dL: < 4%** • **< 54 mg/dL: < 1%** (older/high risk: < 70 → < 1%)
CGM — Time above range	> 180: < 25% • > 250: < 5%
Glycemic variability (CV)	≤ 36%

- **A1C testing:** at least **twice yearly** if at goal and stable • **every 3 months** if therapy changed or not at goal.
- **GMI** (glucose management indicator) = estimated A1C from ≥ 14 days of CGM data.
- **CGM recommended** for all on **insulin** (T1 and T2 basal or multiple daily injections) and considered for T2DM on non-insulin therapy.
- **SMBG**: most important for insulin users — before meals, bedtime, before exercise/driving, when hypoglycemia suspected.
- **Older adults:** healthy < 7–7.5% • complex/intermediate < 8% • very complex/poor health → avoid reliance on A1C; avoid hypoglycemia and symptomatic hyperglycemia.
- **Children/adolescents (T1/T2):** A1C < 7% for most (< 6.5% if safe).
- **Pregnancy:** A1C **< 6%** if no significant hypoglycemia (relax to < 7% if needed).

5. Lifestyle Management

- **Diabetes self-management education and support (DSMES)** at diagnosis, annually, with complications, and at transitions of care.
- **Medical nutrition therapy:** no ideal macronutrient % • emphasize non-starchy vegetables, whole foods, minimize added sugar and refined grains • **carbohydrate counting** for flexible insulin dosing (T1) • replace sugar-sweetened beverages with water.
- **Weight loss ≥ 5%** (overweight/obese T2DM) improves glycemia, lipids, BP • **≥ 10–15%** may produce **remission**.
- **Physical activity: ≥ 150 min/week** moderate aerobic over ≥ 3 days (no > 2 consecutive days without activity) + **resistance 2–3 times/week** • break up sitting every 30 min • check glucose before exercise if on insulin/secretagogues (carbohydrate if < 90–100 mg/dL).
- **Alcohol:** ≤ 1 drink/day women, ≤ 2 men • **risk of delayed hypoglycemia** (inhibits gluconeogenesis) — eat while drinking.
- **Smoking cessation** (all tobacco and e-cigarettes) • sleep 7–9 h • psychosocial screening (depression, diabetes distress, eating disorders).
- **Nonnutritive sweeteners** acceptable as replacement for sugar in the short term.

6. Pharmacologic Treatment of Type 2 Diabetes

ADA approach — choose by comorbidities first, then glycemic needs

Situation	Preferred therapy (independent of A1C / metformin use)
Established ASCVD or high CV risk	**GLP-1 RA** with proven CV benefit (**semaglutide, liraglutide, dulaglutide**) **and/or SGLT2 inhibitor** with proven benefit (**empagliflozin, canagliflozin, dapagliflozin**) • tirzepatide also an option
Heart failure (HFrEF or HFpEF)	**SGLT2 inhibitor** (dapagliflozin, empagliflozin, sotagliflozin) • GLP-1 RA (semaglutide/tirzepatide) for HFpEF symptoms with obesity • **avoid TZDs** and saxagliptin/alogliptin
CKD (eGFR 20–60 and/or albuminuria)	**SGLT2 inhibitor** (start if **eGFR ≥ 20**) • **GLP-1 RA** (semaglutide — FLOW trial) • **finerenone** if albuminuria on max ACEi/ARB • ACEi/ARB
Obesity / weight management	**Tirzepatide** or **semaglutide** (highest weight loss) • other GLP-1 RA • SGLT2i (modest)
MASLD / MASH	**GLP-1 RA**, **tirzepatide**, **pioglitazone**
Minimize hypoglycemia	Metformin, SGLT2i, GLP-1 RA, DPP-4i, TZD (avoid SU, insulin)
Cost is the main concern	**Metformin**, **sulfonylureas**, **pioglitazone**, human insulin
No compelling comorbidity	**Metformin** (+ lifestyle) commonly first-line; agents with high glycemic efficacy and weight benefit

- **Early combination therapy** if A1C is **≥ 1.5% above goal** (e.g., A1C ≥ 8.5%).
- **Start insulin** if **A1C > 10%** or **glucose ≥ 300 mg/dL**, symptomatic hyperglycemia/catabolism (weight loss), or ketosis — can de-escalate later.
- If injectable needed → **GLP-1 RA (or tirzepatide) preferred before insulin** • combine GLP-1 RA with basal insulin (fixed ratio: iDegLira, iGlarLixi).
- **Reassess every 3–6 months**; intensify promptly (avoid clinical inertia) • check adherence and technique first.
- **Do not combine DPP-4 inhibitor with GLP-1 RA/tirzepatide** (same pathway, no added benefit).
- When adding insulin, **stop/reduce sulfonylurea** (hypoglycemia) and keep metformin, GLP-1 RA, SGLT2i.
- Organ benefits of SGLT2i/GLP-1 RA are independent of A1C → add even if A1C at goal.

Comparison of non-insulin classes

Class	A1C ↓	Hypo	Weight	CV / renal effect	Key cautions
Metformin	1–1.5%	No	Neutral/↓	Neutral (possible benefit)	GI, B12, eGFR < 30 CI
SGLT2i	0.5–1%	No	↓	**↓ HF, ↓ CKD progression**, ↓ MACE (empa, cana)	Genital infections, volume, DKA
GLP-1 RA	1–1.5%	No	**↓↓**	**↓ MACE** (lira, sema, dula), ↓ CKD (sema)	GI, MTC/MEN2, pancreatitis
Tirzepatide	~2%	No	**↓↓↓**	Benefit (SURPASS-CVOT)	Same as GLP-1
DPP-4i	0.5–0.8%	No	Neutral	Neutral • ↑ HF (saxa, alo)	Pancreatitis, arthralgia
TZD (pioglitazone)	1–1.5%	No	↑	Possible ASCVD benefit • **↑ HF**	Edema, HF, fractures
Sulfonylureas	1–1.5%	**Yes**	↑	Neutral	Hypoglycemia (elderly, CKD)
Insulin	Highest (no ceiling)	**Yes**	↑	Neutral	Hypoglycemia, weight

7. Non-Insulin Agents — Details

Metformin (biguanide)

- **MOA: ↓ hepatic glucose production** (gluconeogenesis), ↑ insulin sensitivity • no hypoglycemia alone • weight neutral/modest loss • ↓ LDL slightly.
- **Dose:** IR **500 mg once–twice daily with meals** → ↑ 500 mg weekly → max **2,550 mg/day** (effective max ~2,000 mg) • **ER 500–2,000 mg once daily with evening meal**.
- **ADRs: GI** (diarrhea, nausea, metallic taste — ↓ with slow titration, food, ER form), **vitamin B12 deficiency** (check periodically, especially with neuropathy/anemia), **lactic acidosis** (rare, **boxed warning**).
- **Renal dosing: eGFR < 30 → contraindicated** • eGFR 30–45 → **do not start**; if already on it, assess benefit/risk and reduce (max 1,000 mg/day) • check eGFR at least yearly.
- **Iodinated contrast:** hold at the time of/before the procedure if eGFR 30–60, liver disease, alcoholism, HF, or intra-arterial contrast → **restart 48 h later** if renal function stable.
- Also hold in: **acute illness with dehydration**, sepsis, shock, hypoxia, decompensated HF, surgery (day of), excessive alcohol, hepatic impairment.
- **DDIs: cimetidine**, **dolutegravir**, ranolazine, trimethoprim (↓ renal secretion via OCT2/MATE → ↑ metformin) • **topiramate/zonisamide** (carbonic anhydrase inhibitors → acidosis) • alcohol ↑ lactic acidosis risk.

SGLT2 inhibitors ("-gliflozins")

- **MOA:** block **SGLT2** in the proximal tubule → **↑ urinary glucose excretion** (glucosuria) + natriuresis → ↓ BP 3–5 mmHg, ↓ weight 2–3 kg.
- **Doses: empagliflozin 10 → 25 mg** daily (HF/CKD 10 mg) • **dapagliflozin 5 → 10 mg** (HF/CKD 10 mg) • **canagliflozin 100 → 300 mg** before first meal (eGFR < 60: max 100 mg) • **ertugliflozin 5 → 15 mg** • bexagliflozin 20 mg • sotagliflozin (HF) 200 → 400 mg.
- **Kidney function:** glucose-lowering effect falls when eGFR < 45, but **HF/CKD benefits persist** → **start if eGFR ≥ 20**; continue until dialysis • expect a **small initial eGFR dip** (hemodynamic, not a reason to stop).
- **ADRs: genital mycotic infections** (most common, women > men — hygiene), **UTI**/urosepsis/pyelonephritis, **polyuria**, **volume depletion** → **hypotension**, dizziness, **AKI** (especially with diuretics, ACEi/ARB, elderly), **euglycemic DKA** (glucose may be < 250), **Fournier's gangrene** (necrotizing fasciitis of perineum), ↑ LDL, **amputation and fractures (canagliflozin)**.
- **Hold before surgery:** canagliflozin, dapagliflozin, empagliflozin **3 days** • **ertugliflozin 4 days** • also hold during acute illness, fasting, or reduced intake (sick days).
- Euglycemic DKA risk ↑ with: insulin dose reductions, low-carb/ketogenic diet, alcohol, surgery, acute illness, T1DM (not approved for T1DM in US).

- **DDIs:** diuretics (volume), insulin/SU (↓ their dose), **canagliflozin + UGT inducers** (rifampin, phenytoin, ritonavir → ↑ to 300 mg), canagliflozin ↑ digoxin.
- False-positive urine glucose test • ↓ need for diuretics in HF.

GLP-1 receptor agonists and tirzepatide

- **MOA: glucose-dependent ↑ insulin**, **↓ glucagon**, **slow gastric emptying**, ↑ satiety (central) → ↓ postprandial glucose and weight • tirzepatide = **GIP + GLP-1** dual agonist.
- **Doses (T2DM): semaglutide SC (Ozempic) 0.25 mg weekly × 4 weeks → 0.5 → 1 → 2 mg • oral semaglutide (Rybelsus) 3 mg daily × 30 days → 7 → 14 mg** (newer tablets 1.5 → 4 → 9 mg) • **liraglutide (Victoza) 0.6 mg daily × 1 week → 1.2 → 1.8 mg • dulaglutide 0.75 → 1.5 mg weekly** (↑ to 3 → 4.5 mg) • exenatide 5 → 10 mcg BID (within 60 min before morning and evening meals) • **tirzepatide (Mounjaro) 2.5 mg weekly × 4 weeks → ↑ 2.5 mg every 4 weeks to 5–15 mg**.
- **Rybelsus administration:** take on an **empty stomach** with **≤ 4 oz plain water** → wait **30 minutes** before food, drinks, or other oral meds.
- **Boxed warning: thyroid C-cell tumors → CI personal/family history of MTC or MEN 2**.
- **ADRs: nausea, vomiting, diarrhea** (most common — titrate slowly, eat smaller meals, stop when full), constipation, **pancreatitis** (stop; do not restart), **gallbladder disease**, **AKI** (dehydration), injection-site reactions, ↑ HR, **diabetic retinopathy complications** (semaglutide — rapid A1C ↓), aspiration risk with anesthesia, ileus.
- **Avoid in:** severe gastroparesis, history of pancreatitis (caution) • exenatide: avoid CrCl < 30 • others: no renal adjustment.
- **Hypoglycemia only with insulin/SU** → reduce SU or insulin dose (e.g., insulin 10–20%) when starting.
- **DDIs:** delayed gastric emptying → ↓/slowed absorption of oral drugs (narrow therapeutic index: warfarin — monitor INR) • **tirzepatide ↓ oral contraceptive** efficacy (backup 4 weeks after start/each increase) • don't use with DPP-4i or another GLP-1 RA.
- Missed weekly dose: semaglutide within 5 days, dulaglutide within 3 days (72 h), tirzepatide within 4 days — otherwise skip.
- Pens: semaglutide/dulaglutide/tirzepatide pens can be kept at room temp (≤ 30 °C) for 21–56 days per product; liraglutide 30 days after first use.

DPP-4 inhibitors ("-gliptins")

- **MOA:** inhibit DPP-4 → ↑ endogenous GLP-1/GIP (glucose-dependent) • oral, weight neutral, low hypoglycemia, modest A1C ↓ (0.5–0.8%), well tolerated (good for elderly).
- **Doses: sitagliptin 100 mg** daily (eGFR 30–44: 50 mg • < 30: 25 mg) • **saxagliptin 5 mg** (eGFR ≤ 45 or **strong CYP3A4 inhibitor**: 2.5 mg) • **linagliptin 5 mg** (**no renal or hepatic adjustment** — biliary excretion) • **alogliptin 25 mg** (eGFR 30–59: 12.5 mg • < 30: 6.25 mg).
- **ADRs:** nasopharyngitis, headache, URTI, **pancreatitis**, **severe arthralgia**, **bullous pemphigoid**, hypersensitivity (angioedema, SJS), **heart failure hospitalization — saxagliptin and alogliptin**, hepatotoxicity (alogliptin).
- **DDIs:** saxagliptin (CYP3A4 substrate — strong inhibitors ↓ dose, inducers ↓ effect) • linagliptin (P-gp/3A4 inducers like rifampin ↓ effect) • ↑ ACEi angioedema risk.

Thiazolidinediones (TZDs, "glitazones")

- **MOA: PPAR-γ agonists** → ↑ insulin sensitivity in muscle/adipose/liver • **slow onset** (full effect **2–3 months**) • durable effect.
- **Doses: pioglitazone 15–30 mg → max 45 mg daily** (max 15 mg with **gemfibrozil**/strong CYP2C8 inhibitors) • rosiglitazone 4–8 mg (rarely used).
- **Boxed warning: heart failure** (fluid retention) → **CI in NYHA class III–IV HF** • avoid in any symptomatic HF.
- **ADRs: edema** (worse with insulin), **weight gain**, **fractures** (women — distal limbs), **bladder cancer** (pioglitazone — avoid with active/history of bladder cancer), **macular edema**, hepatotoxicity (check **LFTs at baseline**, avoid if ALT > 2.5× ULN), anemia, ovulation in anovulatory women (PCOS → pregnancy).
- **Benefits:** no hypoglycemia, ↓ TG/↑ HDL, **MASH improvement**, possible ↓ stroke/MI (IRIS, PROactive).

Sulfonylureas (insulin secretagogues)

- **MOA:** close K-ATP channels on β-cells → **↑ insulin secretion regardless of glucose** → **hypoglycemia** • rapid A1C ↓ (1–1.5%) but **secondary failure** over time.
- **Doses: glipizide 2.5–5 mg daily** (IR 30 min before breakfast) → max 20 mg/day (XL 20 mg; label 40 mg IR) • **glimepiride 1–2 mg daily** with breakfast → max 8 mg • **glyburide 1.25–5 mg** → max 20 mg (micronized 12 mg).
- **Glyburide: avoid in elderly and CKD** (active renal metabolites → prolonged hypoglycemia; **Beers**) • **glipizide preferred in CKD/elderly** (hepatic metabolism, inactive metabolites).
- **ADRs: hypoglycemia** (elderly, renal/hepatic impairment, skipped meals, alcohol), **weight gain** (~2 kg), rash, hemolytic anemia with **G6PD deficiency**, photosensitivity • sulfonamide allergy cross-reactivity is low (caution only).
- **DDIs: CYP2C9 inhibitors** (**fluconazole**, amiodarone, **sulfamethoxazole/TMP**, gemfibrozil) → ↑ hypoglycemia • **fluoroquinolones** (dysglycemia) • **beta-blockers mask hypoglycemia symptoms** (except sweating) • alcohol (disulfiram-like reaction with chlorpropamide).

Meglitinides (glinides)

- **Repaglinide** 0.5–4 mg (max 16 mg/day) and **nateglinide** 60–120 mg TID **15–30 min before meals** → short-acting, target postprandial glucose.
- **"Skip a meal, skip a dose; add a meal, add a dose"** • hypoglycemia (less than SU), weight gain • TID dosing.
- **Repaglinide + gemfibrozil = contraindicated** (CYP2C8 → severe hypoglycemia) • clopidogrel (2C8 inhibitor) ↑ repaglinide • useful in CKD (repaglinide).

Other agents

- **Alpha-glucosidase inhibitors** (**acarbose** 25 mg TID **with the first bite of each meal** → 50–100 mg TID; miglitol): delay carbohydrate absorption → ↓ postprandial • ADRs: **flatulence, bloating, diarrhea** • **treat hypoglycemia with glucose (dextrose) — not sucrose** • CI: IBD, bowel obstruction, cirrhosis, SCr > 2 • ↑ LFTs (acarbose).
- **Colesevelam** 3.75 g/day (bile acid sequestrant): ↓ A1C 0.5% + ↓ LDL • constipation, **↑ TG** (CI TG > 500) • separate **levothyroxine, OCs, glyburide, phenytoin by 4 h**.
- **Bromocriptine QR** 0.8 → 4.8 mg within 2 h of waking (with food): dopamine agonist • nausea, orthostasis, somnolence • CI with syncopal migraine, postpartum, breastfeeding.
- **Pramlintide** (amylin analog; T1 and T2 on prandial insulin): T1 **15 mcg** → 60 mcg; T2 **60 mcg** → 120 mcg SC before major meals • **boxed: severe hypoglycemia → ↓ prandial insulin by 50%** when starting • nausea • do not mix with insulin • CI gastroparesis, hypoglycemia unawareness.

8. Insulin

Insulin products

Type	Products	Onset	Peak	Duration	Notes
Rapid-acting	**Lispro** (Humalog, Admelog), **aspart** (Novolog), **glulisine** (Apidra)	15 min	1–2 h	3–5 h	Inject **0–15 min before meals** • pumps • can be IV
Ultra-rapid	**Faster aspart** (Fiasp), **lispro-aabc** (Lyumjev)	2–5 min	1–2 h	3–5 h	At start of meal (or within 20 min after)
Inhaled	**Afrezza** (Technosphere)	~12 min	~1 h	2–3 h	**CI asthma, COPD, smokers** • bronchospasm (boxed) • spirometry before and periodically
Short-acting	**Regular** (Humulin R, Novolin R)	30 min	2–3 h	5–8 h	**30 min before meals • only insulin for IV** drips (with rapid analogs) • DKA
Concentrated regular	**U-500 regular**	30 min	4–8 h	Up to 24 h	Severe insulin resistance (> 200 units/day) • **dedicated U-500 pen/syringe**
Intermediate	**NPH** (Humulin N, Novolin N)	1–2 h	**4–12 h**	12–18 h	**Cloudy** (roll gently) • BID • risk of nocturnal hypoglycemia • steroid hyperglycemia
Long-acting	**Glargine U-100** (Lantus, Basaglar, Semglee, Rezvoglar), **detemir** (Levemir — discontinued US)	1–2 h	**No (flat)**	~24 h (detemir 12–24 h)	Once daily (detemir may need BID) • **do not mix** with other insulins
Ultra-long	**Glargine U-300** (Toujeo), **degludec** U-100/U-200 (Tresiba)	6 h / 1 h	No	> 24 h (degludec ~42 h)	Less nocturnal hypoglycemia • degludec flexible timing (≥ 8 h between doses)
Premixed	NPH/regular **70/30** • lispro mix 75/25, 50/50 • aspart mix 70/30	Varies	Dual	10–16 h	BID before breakfast and dinner • fixed ratio, less flexible, consistent meals needed

- **Mixing:** only NPH with rapid/regular — **"clear before cloudy"** (draw regular/rapid first, then NPH) • **never mix glargine, detemir, degludec**.
- **Storage:** unopened → **refrigerate** (2–8 °C) until expiration (do not freeze) • **in use → room temperature ~28 days** for most vials/pens (**degludec and Toujeo 56 days**, detemir 42 days, NPH pens ~14 days, premixed pens 10–14 days).
- **Injection sites:** abdomen (fastest absorption) > arm > thigh > buttock • **rotate sites** to prevent **lipohypertrophy** (erratic absorption) • hold pen needle **10 seconds** after injection • prime pen with 2 units.
- **Concentrated insulins:** U-200 lispro/degludec, U-300 glargine, U-500 regular → **pens only (except U-500 has its own syringe)** — never draw from pens with syringes.
- Insulin **ADRs: hypoglycemia**, **weight gain**, lipohypertrophy, injection-site reactions, **hypokalemia** (IV; shifts K into cells), edema (with TZDs).

Insulin in type 1 diabetes

- **Physiologic regimen: multiple daily injections (MDI)** (basal + prandial) or **insulin pump (CSII)** • **automated insulin delivery (AID, hybrid closed loop)** preferred when possible.
- **Total daily dose (TDD): 0.4–1.0 units/kg/day** (usual start **0.5 units/kg/day**; honeymoon phase lower ~0.2–0.4; puberty, illness, pregnancy higher).
- **Split: ~50% basal** (once daily long-acting) + **~50% prandial** divided among meals (rapid-acting).
- **Insulin-to-carbohydrate ratio (ICR): 500 ÷ TDD** = grams of carbohydrate covered by 1 unit (450 rule for regular).
- **Correction (sensitivity) factor (ISF): 1,800 ÷ TDD** (rapid-acting) or **1,500 ÷ TDD** (regular) = mg/dL drop per 1 unit.
- **Correction dose** = (current BG − target BG) ÷ ISF • **meal dose** = carbs ÷ ICR • total prandial = meal + correction.

- **Pump:** uses rapid-acting only (no basal insulin) → **pump failure → DKA within hours** → backup injections.

Worked example: 70-kg T1DM → TDD = 0.5 × 70 = **35 units** → basal **~17 units** glargine daily • prandial ~6 units per meal • ICR = 500/35 ≈ **1 unit : 14 g carbs** • ISF = 1,800/35 ≈ **1 unit lowers 50 mg/dL**. Meal with 70 g carbs and BG 230 (target 130): 70/14 = 5 units + (230 − 130)/50 = 2 units → **7 units** rapid-acting.

Insulin in type 2 diabetes

- **Start basal insulin: 10 units/day** or **0.1–0.2 units/kg/day** (often at bedtime for NPH, any consistent time for glargine/degludec).
- **Titrate** by **2 units every 3 days** (or 10–15%, or 2–4 units once–twice weekly) until **fasting glucose at goal (80–130)**.
- **Hypoglycemia** without clear cause → **↓ dose by 4 units or 10–20%**.
- **Overbasalization** — signs: **basal dose > 0.5 units/kg/day**, bedtime-to-morning glucose difference ≥ 50 mg/dL, hypoglycemia, high variability → **add GLP-1 RA** (preferred) **or prandial insulin** rather than more basal.
- **Add prandial insulin:** start **4 units or 10% of basal** before the **largest meal** (basal-plus) → titrate 1–2 units or 10–15% twice weekly → basal-bolus (reduce basal by the same amount if A1C < 8%).
- **Switching:** NPH BID → glargine/degludec once daily = **80% of total NPH dose** • glargine U-300 → U-100 = 80% • U-100 → U-300 = 1:1 (may need 10–18% more) • detemir → glargine 1:1.
- **Premixed 70/30**: 2/3 TDD before breakfast, 1/3 before dinner (needs consistent meals).
- Continue metformin, GLP-1 RA, SGLT2i with insulin • **stop or reduce SU** (and avoid TZD with insulin — edema/HF).

Factors affecting insulin needs

- **↑ insulin needs:** infection/illness, stress, surgery, **glucocorticoids**, puberty, pregnancy (2nd/3rd trimester), weight gain, **dawn phenomenon** (GH/cortisol early morning).
- **↓ insulin needs:** exercise, weight loss, **CKD** (↓ insulin clearance), hepatic failure, adrenal insufficiency, hypothyroidism, early pregnancy/**after delivery**, gastroparesis.
- **Somogyi effect** (rebound hyperglycemia after nocturnal hypoglycemia) → check 3 AM glucose (CGM) → **↓ bedtime basal**; dawn phenomenon → ↑/shift basal.

9. Hypoglycemia

Level	Definition
Level 1	Glucose **< 70 mg/dL and ≥ 54 mg/dL** (alert value)
Level 2	Glucose **< 54 mg/dL** (clinically significant)
Level 3 (severe)	**Altered mental/physical status requiring assistance** from another person (any glucose)

- **Symptoms — autonomic (adrenergic): tremor, palpitations, sweating**, anxiety, hunger • **neuroglycopenic:** confusion, dizziness, blurred vision, seizures, coma.
- **Beta-blockers mask** tremor and tachycardia (**not sweating**) and delay recovery • **hypoglycemia unawareness** (repeated episodes, long duration, autonomic neuropathy) → **raise glucose targets** and strictly avoid hypoglycemia for several weeks to restore awareness; CGM with alarms.
- **Causes:** insulin or SU/glinide excess, missed/delayed meals, **exercise**, **alcohol**, **CKD**, weight loss, drug interactions (fluoroquinolones, fluconazole with SU), pentamidine, quinine.

Treatment — "Rule of 15" (conscious patient): 15–20 g of fast-acting glucose (pure glucose preferred) → recheck in **15 minutes** → repeat if still < 70 → once normal, eat a **meal or snack** to prevent recurrence. **15 g carbohydrate =** 4 glucose tablets (4 g each), 4 oz (½ cup) juice or regular soda, 1 tablespoon sugar/honey, glucose gel. **Avoid fat/protein-rich foods** (chocolate, milk — slow absorption; protein ↑ insulin). **Acarbose/miglitol users → glucose only** (not sucrose). **Severe/unconscious: glucagon** (**1 mg IM/SC**, **nasal glucagon 3 mg**, **dasiglucagon 0.6 mg SC**) — family members trained; in hospital **IV dextrose (D50 25 g / D10)**. Place on the side (vomiting). Do not give oral food to unconscious patients.

- **Prescribe glucagon** for all at risk of level 2–3 hypoglycemia (insulin users, high-risk SU users) • glucagon less effective in alcohol intoxication/starvation (depleted glycogen).
- **Sulfonylurea-induced hypoglycemia** may be prolonged → observe; **octreotide** for refractory cases.
- Check glucose before driving (≥ 90) and carry glucose.

10. Hyperglycemic Crises — DKA and HHS

Feature	DKA	HHS
Typical patient	**T1DM** (also T2DM, SGLT2i users)	**Older T2DM**
Onset	Hours to 1 day	Days to weeks
Glucose	**≥ 200 mg/dL** (or known diabetes) • can be normal (euglycemic DKA)	**≥ 600 mg/dL**
pH / HCO3	**pH < 7.3** and/or **HCO3 < 18**	pH ≥ 7.3, HCO3 ≥ 15
Ketones	**β-hydroxybutyrate ≥ 3 mmol/L** (or urine ketones ≥ 2+)	Small/none (< 3)
Osmolality	Variable	**Effective osmolality > 300** (usually > 320) mOsm/kg
Anion gap	**Elevated** (> 10–12)	Variable
Mental status	Alert to drowsy	**Stupor/coma** common
Clinical	**Kussmaul respirations**, **fruity breath**, abdominal pain, N/V	Profound **dehydration** (~9 L deficit)

- **Precipitants ("I's"): infection** (most common), **insulin omission**/pump failure, new-onset diabetes, infarction (MI, stroke), pancreatitis, **drugs** (**glucocorticoids**, **SGLT2i**, **atypical antipsychotics**, thiazides, sympathomimetics, **immune checkpoint inhibitors**), cocaine, alcohol, pregnancy.
- **Anion gap** = Na − (Cl + HCO3) • **corrected Na** = Na + 1.6 × (glucose − 100)/100 • **effective osmolality** = 2 × Na + glucose/18.

Treatment (ADA 2024 consensus)

Step	What to do
1. Fluids (first priority)	**0.9% NaCl or balanced crystalloid (LR, Plasma-Lyte) 1–1.5 L (15–20 mL/kg) in the 1st hour** → then 250–500 mL/h based on volume status and corrected Na (0.45% NaCl if corrected Na normal/high) • **add dextrose (D5 or D10) when glucose < 250 mg/dL (DKA)** or **< 300 (HHS)**
2. Check K before insulin	**K < 3.5 → hold insulin**, give K **10–20 mEq/h** until K ≥ 3.5 • **K 3.5–5.0 → add 10–20 mEq/h** (keep 4–5) • **K > 5.0 → no K**, recheck every 2 h
3. Insulin	**Regular insulin IV infusion 0.1 units/kg/h** (± 0.1 units/kg bolus) or 0.14 units/kg/h without bolus • goal ↓ glucose **50–75 mg/dL/h** • when glucose < 250 → ↓ to **0.05 units/kg/h** + dextrose; keep glucose 150–250 until resolution (HHS 250–300) • mild–moderate DKA: SC rapid-acting analog every 1–2 h is an option
4. Bicarbonate	**Only if pH < 7.0** (100 mEq in 400 mL water + 20 mEq KCl over 2 h)
5. Phosphate	Only if **< 1.0 mg/dL** or cardiac dysfunction, respiratory depression, hemolytic anemia (K-phos 20–30 mEq)
6. Treat precipitant	Antibiotics, etc. • monitor glucose hourly, electrolytes/pH/ketones every 2–4 h

- **Resolution of DKA: β-hydroxybutyrate < 0.6 mmol/L** and **venous pH ≥ 7.3** or **HCO3 ≥ 18** (glucose does not need to be normal) • **HHS resolved:** osmolality < 300, glucose < 250, recovery of mental status.
- **Transition to SC insulin:** give **basal insulin SC 1–2 h BEFORE stopping the IV insulin** (overlap prevents rebound DKA) • new patients: TDD 0.5–0.6 units/kg/day (50% basal) • early low-dose basal during the infusion is acceptable.
- **Complications: hypoglycemia**, **hypokalemia** (most common, insulin shifts K into cells — K falls even though total body K is depleted), **cerebral edema** (children — avoid rapid osmolality decline), hyperchloremic non-gap acidosis (excess NaCl), fluid overload, VTE (HHS).
- Initial K may be normal/high despite total body deficit (acidosis + insulin deficiency shift K out of cells).
- **Euglycemic DKA (SGLT2i):** glucose < 250 with ketoacidosis → stop SGLT2i, treat with fluids + **dextrose early** + insulin.

11. Cardiovascular Risk Management

Hypertension

- **BP goal < 130/80 mmHg** (if safely achievable) • measure every visit • lifestyle (DASH, Na < 2,300 mg, weight loss).
- **First-line: ACEi, ARB, thiazide-like diuretic, or dihydropyridine CCB** • ≥ 150/90 → start **2 drugs** (single-pill combo).
- **Albuminuria (UACR ≥ 30 mg/g) or CAD → ACEi or ARB** at maximum tolerated dose (monitor K and SCr within 1–2 weeks) • **never ACEi + ARB** or with direct renin inhibitor.
- Resistant HTN → add **MRA** (spironolactone) • **pregnancy: ACEi/ARB/statins contraindicated** → labetalol, nifedipine.

Lipids — statin therapy

Patient	Recommendation
Age 40–75 without ASCVD	**Moderate-intensity statin** (at least)
Age 40–75 with ≥ 1 ASCVD risk factor	**High-intensity statin** → goal LDL ↓ ≥ 50% and **LDL < 70 mg/dL** (consider ezetimibe or PCSK9i)
Established ASCVD (any age)	**High-intensity statin** → **LDL < 55 mg/dL** • add **ezetimibe** or **PCSK9 inhibitor** (or bempedoic acid if statin-intolerant)
Age 20–39 with risk factors / > 75	Reasonable to start/continue statin
TG ≥ 150 + ASCVD/risk factors on statin	**Icosapent ethyl** 2 g BID • TG ≥ 500 → fibrate/omega-3, check secondary causes

- **High-intensity:** atorvastatin 40–80 mg, rosuvastatin 20–40 mg • **moderate:** atorvastatin 10–20, rosuvastatin 5–10, simvastatin 20–40, pravastatin 40–80 mg.
- Statins slightly ↑ glucose/A1C — benefit far outweighs risk.
- Lipid panel at diagnosis/start, 4–12 weeks after starting, then annually.

Antiplatelet therapy

- **Aspirin 75–162 mg/day** for **secondary prevention** (ASCVD) • clopidogrel 75 mg if aspirin allergy • DAPT after ACS per cardiology.
- **Primary prevention:** consider aspirin only in **high CV risk** after discussing bleeding risk (generally not > 70 years).
- **Rivaroxaban 2.5 mg BID + aspirin** for stable CAD/PAD with low bleeding risk.

Heart failure

- Screen with BNP/NT-proBNP • **SGLT2i** for all HF (HFrEF and HFpEF) • HFrEF: ARNI/ACEi/ARB, beta-blocker, MRA, SGLT2i • **avoid TZDs**, saxagliptin, alogliptin.

12. Microvascular Complications and Foot Care

Chronic kidney disease (diabetic kidney disease)

- **Screen yearly: UACR** (spot urine albumin-to-creatinine ratio) + **eGFR** — T1DM starting **5 years after diagnosis**, **T2DM at diagnosis**.
- **Albuminuria:** UACR **< 30** normal • **30–299** moderately increased (A2, "microalbuminuria") • **≥ 300** severely increased (A3) • confirm with **2 of 3 samples** over 3–6 months (exercise, infection, fever, HF, marked hyperglycemia, menstruation, HTN can cause false elevations).
- **Treatment pillars: glycemic + BP control • ACEi/ARB** (if albuminuria or HTN; max tolerated dose; not for primary prevention in normotensive normal UACR) • **SGLT2i** (eGFR ≥ 20) • **finerenone** (nonsteroidal MRA: UACR ≥ 30, eGFR ≥ 25, K ≤ 5.0 — monitor K) • **GLP-1 RA** (semaglutide) • statin.
- Protein intake **0.8 g/kg/day** (non-dialysis) • monitor K and SCr with ACEi/ARB/finerenone • refer to nephrology if eGFR < 30 or rapidly progressing.
- **Renal dose adjustments:** metformin, sitagliptin/saxagliptin/alogliptin, glyburide (avoid), exenatide, SGLT2i glycemic efficacy ↓, insulin requirements ↓.

Retinopathy

- **Dilated eye exam: T1DM within 5 years** of diagnosis • **T2DM at diagnosis** • then every **1–2 years** if no retinopathy and well controlled; **yearly** or more if retinopathy present.
- **Pregnancy:** exam before conception/1st trimester, every trimester, and 1 year postpartum (retinopathy can worsen rapidly).
- **Treatment:** glucose, BP, lipid control • **anti-VEGF intravitreal injections** (ranibizumab, aflibercept, bevacizumab) for diabetic macular edema and proliferative retinopathy • panretinal laser photocoagulation • fenofibrate may slow progression.
- Aspirin does not increase retinal hemorrhage risk • rapid A1C reduction (semaglutide, insulin) may cause early worsening.

Neuropathy

- **Distal symmetric polyneuropathy** (most common): "stocking-glove" numbness, burning pain, tingling, loss of protective sensation → **foot ulcers**.
- **Screen yearly:** T2DM at diagnosis, T1DM 5 years after diagnosis • **10-g monofilament** + pinprick/temperature, vibration (128-Hz tuning fork), ankle reflexes.
- **Painful neuropathy — first-line: gabapentinoids** (**pregabalin** 50 mg TID → 300–600 mg/day; gabapentin 300 → 1,800–3,600 mg/day), **SNRIs** (**duloxetine** 60 mg daily; venlafaxine), **TCAs** (amitriptyline 10–75 mg at bedtime — anticholinergic; avoid in elderly/CVD), **sodium channel blockers** (carbamazepine, oxcarbazepine, lacosamide) • topical **capsaicin 8% patch**, lidocaine • **avoid opioids** (including tramadol/tapentadol) as first-line.
- Gabapentinoids: sedation, dizziness, **edema**, weight gain, renal dose adjustment, respiratory depression with opioids • duloxetine: nausea, avoid CrCl < 30 and hepatic disease.
- Check **vitamin B12** in patients on metformin with neuropathy.

Autonomic neuropathy

- **Gastroparesis:** early satiety, nausea, erratic glucose (hypoglycemia after meals) → small low-fat low-fiber meals; **metoclopramide** (**≤ 12 weeks**, boxed tardive dyskinesia), erythromycin (short-term) • stop GLP-1 RA, pramlintide, opioids, anticholinergics.
- **Cardiovascular:** resting tachycardia, **orthostatic hypotension** (midodrine, droxidopa, fludrocortisone), silent MI • **hypoglycemia unawareness**.
- **Genitourinary: erectile dysfunction** (**PDE5 inhibitors** — CI with nitrates), neurogenic bladder, retrograde ejaculation • sudomotor (dry skin).

Foot care

- **Comprehensive foot exam yearly** (inspection, pulses, monofilament, PAD screening with ABI if symptoms) • **inspect feet at every visit** if high risk.

- **Patient education:** inspect feet daily (mirror), wash and dry well (between toes), moisturize (**not between toes**), **never walk barefoot**, proper shoes/socks, check water temperature, cut nails straight across, no heating pads, see podiatrist for calluses/corns.
- **Diabetic foot infection:** mild → oral (cephalexin, amoxicillin-clavulanate, clindamycin, doxycycline ± MRSA coverage) • moderate–severe → IV broad-spectrum (vancomycin + piperacillin-tazobactam) • osteomyelitis (probe-to-bone) • offloading, debridement.

13. Other Preventive Care

- **Vaccines: influenza** yearly, **pneumococcal** (PCV20 or PCV21, or PCV15 → PPSV23), **hepatitis B** (age 19–59; consider ≥ 60), **COVID-19**, **zoster** (recombinant, ≥ 50), Tdap/Td, **RSV** (older adults per age/risk recommendations), HPV through age 26.
- Dental exams (periodontal disease) • screen for depression, diabetes distress, cognitive impairment (older adults), **MASLD** (FIB-4), sleep apnea, hearing loss, osteoporosis (TZDs, T1DM).
- **Annual checklist:** A1C (2–4×/year), BP each visit, lipids, UACR + eGFR, dilated eye exam, comprehensive foot exam, B12 (metformin), vaccines, weight/BMI, dental, psychosocial.

14. Special Populations

Pregnancy — preexisting diabetes and GDM

- **Preconception:** A1C **< 6.5%** before conception (↓ congenital anomalies and miscarriage) • **folic acid** (400 mcg; higher in some guidelines) • effective contraception until ready • stop **ACEi/ARB, statins**, most non-insulin agents (GLP-1 RA: stop 2 months before for semaglutide).
- **GDM screening at 24–28 weeks** (earlier, < 15 weeks, if risk factors — test for preexisting diabetes):
- **One-step:** 75-g OGTT fasting — diagnose if **any** value ≥: **fasting 92**, **1-h 180**, **2-h 153 mg/dL**.
- **Two-step:** 50-g glucose challenge (nonfasting) → if 1-h **≥ 130–140** → 100-g 3-h OGTT (Carpenter–Coustan: fasting 95, 1-h 180, 2-h 155, 3-h 140) — **≥ 2 abnormal values** = GDM.
- **Glucose goals in pregnancy: fasting < 95**, **1-h postprandial < 140**, **2-h postprandial < 120 mg/dL** • CGM TIR (63–140) > 70% for T1DM.
- **Treatment:** lifestyle (MNT, activity) first in GDM (70–85% controlled) → **insulin is preferred** (does not cross placenta) • **metformin and glyburide cross the placenta** → not first-line (metformin: possible childhood weight effects; glyburide: neonatal hypoglycemia, macrosomia).
- Insulin needs: ↓ in 1st trimester (hypoglycemia risk), ↑ **markedly in 2nd–3rd trimester** (placental hormones) → **drop sharply after delivery** (return to pre-pregnancy or less).
- **Low-dose aspirin 100–150 mg/day** (often 81–162 mg) from **12–16 weeks** in T1/T2DM to ↓ **preeclampsia**.
- BP goal in pregnancy ~110–135/85 • **labetalol, nifedipine, methyldopa**.
- **Postpartum GDM: 75-g OGTT at 4–12 weeks** postpartum (not A1C) → then screening **every 1–3 years lifelong** (up to 50–70% develop T2DM) • breastfeeding encouraged.
- **Complications of poorly controlled diabetes in pregnancy:** congenital anomalies (cardiac, neural tube, caudal regression), miscarriage, **macrosomia**, shoulder dystocia, neonatal hypoglycemia, preeclampsia, stillbirth.
- **Breastfeeding:** insulin, metformin, glyburide/glipizide acceptable.

Children and adolescents

- **T1DM:** most common in children • A1C < 7% • insulin pumps/AID, CGM • screen for celiac disease and autoimmune thyroid disease (TSH) at diagnosis and periodically • **cerebral edema** in DKA.
- **T2DM in youth: metformin** first-line (if A1C < 8.5% and no ketosis) • insulin if A1C ≥ 8.5% or ketosis/DKA • approved: **GLP-1 RA** (liraglutide, dulaglutide, exenatide ER, semaglutide/tirzepatide approvals expanding), **empagliflozin and dapagliflozin (≥ 10 years)** • SU and TZDs not recommended.

Older adults (≥ 65)

- Individualize goals (healthy < 7–7.5%, complex < 8%) • **avoid hypoglycemia** (falls, cognitive impairment, CV events) • **deintensify** when possible.

- **Beers criteria: avoid glyburide and glimepiride** (prolonged hypoglycemia; glipizide preferred if SU needed), **avoid sliding-scale insulin alone** (hyperglycemia/hypoglycemia without basal), **SGLT2i with caution** (volume depletion, UTI, genital infections), TZDs in HF.
- Prefer: DPP-4i, low-dose basal insulin (morning), GLP-1 RA (watch weight/muscle loss), SGLT2i (watch volume).

Hospital care

- **Target glucose 140–180 mg/dL** for most (critically and noncritically ill) • 100–180 acceptable in selected patients • start insulin if persistently ≥ 180.
- **Critically ill: IV insulin infusion** (regular insulin) with validated protocol.
- **Non-critically ill: basal-bolus** (eating) or **basal + correction** (NPO/poor intake) • **sliding-scale insulin alone is discouraged** • starting TDD 0.3–0.5 units/kg (0.1–0.25 in elderly, CKD, insulin-naive) • half basal, half prandial.
- Non-insulin agents usually **held** in hospital (DPP-4i may be used in mild hyperglycemia).
- **Hypoglycemia protocol** in every hospital • A1C on admission if none in past 3 months • reduce basal 10–20% if hypoglycemia.
- **Perioperative:** goal 100–180 mg/dL • **metformin hold day of surgery** • **SGLT2i hold 3–4 days** before • SU/glinides hold day of surgery • give **~75–80% of long-acting basal** (glargine) the night before/morning (NPH 50% morning dose) • follow anesthesia guidance for GLP-1 RA (risk of aspiration).
- **Enteral/parenteral nutrition:** basal + scheduled insulin covering feeds (regular every 6 h or NPH every 8–12 h) • **regular insulin in PN bag**.
- **Glucocorticoid-induced hyperglycemia:** afternoon/evening hyperglycemia with morning prednisone → **NPH given with the morning steroid** (≈ 0.1 units/kg per 10 mg prednisone, max 0.4 units/kg) • taper insulin with steroid taper.

Sick-day management

- **Never stop basal insulin** (T1DM — DKA risk) even if not eating • may need **more** insulin.
- Check glucose every **2–4 h** • **check ketones** (T1DM) if glucose > 250 or illness • drink fluids (sugar-free if high; sugar-containing if unable to eat) • 45–50 g carbs every 3–4 h.
- **Hold ("SADMANS") if dehydrated/vomiting: S**ulfonylureas, **A**CEi, **D**iuretics, **M**etformin, **A**RBs, **N**SAIDs, **S**GLT2i — restart after 24–48 h eating/drinking.
- Seek care: persistent vomiting, moderate–large ketones, glucose > 300 for > 24 h, confusion, inability to keep fluids.

15. Drugs Affecting Glucose

Effect	Drugs
Hyperglycemia	**Glucocorticoids** (most common), **atypical antipsychotics** (**olanzapine, clozapine** > quetiapine, risperidone), **thiazides** (high dose), **non-vasodilating beta-blockers**, **statins** (small), **niacin**, **calcineurin inhibitors** (**tacrolimus** > cyclosporine), sirolimus, **protease inhibitors**, **fluoroquinolones** (dysglycemia), **immune checkpoint inhibitors** (autoimmune T1DM/DKA), **pentamidine** (late), sympathomimetics (epinephrine, albuterol high dose), **octreotide**, diazoxide, growth hormone, oral contraceptives (high estrogen), L-asparaginase, megestrol
Hypoglycemia	Insulin, sulfonylureas, glinides • **alcohol** • **fluoroquinolones** (especially with SU, elderly) • **pentamidine** (early) • **quinine**, quinidine • **SMX/TMP, fluconazole** (with SU) • salicylates (high dose) • ACE inhibitors (↑ sensitivity) • tramadol • disopyramide
Mask hypoglycemia	**Beta-blockers** (nonselective > selective) — mask tremor, palpitations; **sweating remains**

16. Non-Insulin Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Metformin	**500 mg daily–BID with meals** → max 2,550 mg (ER 2,000 mg daily)	**GI**, **B12 deficiency**, lactic acidosis (boxed)	**CI eGFR < 30** • don't start 30–45 • hold for contrast/surgery/acute illness • cimetidine, dolutegravir ↑ • topiramate (acidosis)
Empagliflozin	10 → 25 mg daily (HF/CKD 10 mg)	**Genital infections, UTI**, volume depletion, **euglycemic DKA**, Fournier's	↓ HF, CKD, CV death • start eGFR ≥ 20 • hold 3 days pre-op
Dapagliflozin	5 → 10 mg daily (HF/CKD 10 mg)	Same	↓ HF and CKD progression • hold 3 days pre-op
Canagliflozin	100 → 300 mg before 1st meal (eGFR < 60: 100 mg)	Same + **amputation, fractures**, hyperkalemia	**UGT inducers** → 300 mg • ↑ digoxin • ↓ CKD progression
Ertugliflozin	5 → 15 mg daily	Same	**Hold 4 days pre-op** • CV neutral
Semaglutide SC (Ozempic)	**0.25 mg weekly × 4 wk** → 0.5 → 1 → 2 mg	**N/V/D**, pancreatitis, gallbladder, **retinopathy complications**	**Boxed MTC/MEN2** • ↓ MACE, ↓ CKD • ↓ SU/insulin doses
Semaglutide PO (Rybelsus)	**3 mg × 30 days** → 7 → 14 mg	Same	**Empty stomach, ≤ 4 oz water, wait 30 min**
Liraglutide (Victoza)	**0.6 mg daily × 1 wk** → 1.2 → 1.8 mg	GI, pancreatitis, gallbladder	Boxed MTC • ↓ MACE • ≥ 10 years
Dulaglutide	**0.75 → 1.5 mg weekly** (max 4.5 mg)	GI, pancreatitis	Boxed MTC • ↓ MACE
Exenatide	5 → 10 mcg BID within 60 min before meals	GI, pancreatitis	**Avoid CrCl < 30**
Tirzepatide (Mounjaro)	**2.5 mg weekly × 4 wk** → ↑ 2.5 mg q4wk → 5–15 mg	GI, pancreatitis, gallbladder	Boxed MTC • greatest A1C/weight ↓ • **↓ oral contraceptives**
Sitagliptin	**100 mg daily** (eGFR 30–44: 50 • < 30: 25)	Pancreatitis, arthralgia, bullous pemphigoid	Weight neutral • not with GLP-1 RA
Saxagliptin	5 mg (eGFR ≤ 45 or strong 3A4 inhibitor: 2.5)	Same + **HF**	**CYP3A4**
Linagliptin	**5 mg daily**	Same	**No renal adjustment** • inducers ↓
Alogliptin	25 mg (eGFR 30–59: 12.5 • < 30: 6.25)	Same + **HF**, hepatotoxicity	
Pioglitazone	**15–30 → 45 mg daily**	**Edema, HF (boxed)**, weight gain, fractures, bladder cancer	CI NYHA III–IV • LFTs • **gemfibrozil → max 15 mg** • slow onset • MASH
Glipizide	**2.5–5 mg daily** 30 min before breakfast → max 20 mg	**Hypoglycemia**, weight gain	**Preferred SU in CKD/elderly** • CYP2C9 inhibitors ↑
Glimepiride	**1–2 mg daily** → max 8 mg	Hypoglycemia, weight gain	Beers: avoid in elderly
Glyburide	1.25–5 mg → max 20 mg	**Prolonged hypoglycemia**	**Avoid elderly/CKD** • crosses placenta
Repaglinide	0.5–4 mg 15–30 min before meals (max 16 mg/day)	Hypoglycemia, weight gain	**CI with gemfibrozil** • clopidogrel ↑ • skip meal, skip dose
Nateglinide	60–120 mg TID before meals	Hypoglycemia	Skip meal, skip dose

Drug	Dose	Key ADRs	Key notes / interactions
Acarbose	25 mg TID with first bite → 50–100 mg TID	**Flatulence**, diarrhea, ↑ LFTs	**Hypoglycemia → glucose only** • CI IBD, cirrhosis
Colesevelam	3.75 g/day	Constipation, **↑ TG**	Separate levothyroxine, OCs, glyburide 4 h
Bromocriptine QR	0.8 → 4.8 mg in morning	Nausea, orthostasis	Not with antipsychotics
Pramlintide	T1: 15 → 60 mcg • T2: 60 → 120 mcg before meals	**Severe hypoglycemia (boxed)**, nausea	**↓ prandial insulin 50%** • don't mix with insulin
Teplizumab	IV daily × 14 days	**Lymphopenia**, CRS, rash	Delays stage 3 T1DM (stage 2, age ≥ 8)

17. Insulin and Complication-Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Basal insulin (glargine, degludec)	T2: **10 units or 0.1–0.2 units/kg** → ↑ 2 units q3d • T1: ~50% of TDD	**Hypoglycemia**, weight gain, lipohypertrophy	> 0.5 units/kg = overbasalization • **don't mix** • degludec/Toujeo 56 days
NPH	BID or bedtime • steroids: AM with prednisone	Nocturnal hypoglycemia	Cloudy • clear before cloudy • NPH→glargine 80%
Rapid-acting (lispro, aspart, glulisine)	0–15 min before meals • 4 units or 10% basal start • ICR 500/TDD • ISF 1800/TDD	Hypoglycemia	Pumps • IV possible
Regular insulin	30 min before meals • DKA **0.1 units/kg/h IV**	Hypoglycemia, **hypokalemia**	Only IV insulin (with rapid) • PN bag • U-500
Inhaled insulin (Afrezza)	At start of meal	**Bronchospasm (boxed)**, cough	CI asthma, COPD, smokers • spirometry
Glucagon	**1 mg IM/SC/IV** • nasal 3 mg • dasiglucagon 0.6 mg	N/V	Severe hypoglycemia • less effective in alcohol/starvation
Dextrose 50%	25 g (50 mL) IV	Phlebitis	Hospital severe hypoglycemia
Potassium chloride (DKA)	10–20 mEq/h IV (K < 5.0)	Hyperkalemia, phlebitis	**Hold insulin if K < 3.5**
Sodium bicarbonate (DKA)	100 mEq over 2 h if pH < 7.0	Hypokalemia, alkalosis	Not routine
ACEi / ARB (lisinopril, losartan)	Max tolerated dose	**Hyperkalemia**, ↑ SCr, cough/angioedema (ACEi)	Albuminuria • **CI pregnancy** • not together
Finerenone	10–20 mg daily (by eGFR, K)	**Hyperkalemia**	**Strong CYP3A4 inhibitors CI** • UACR ≥ 30
Atorvastatin / rosuvastatin	High: 40–80 / 20–40 mg • moderate: 10–20 / 5–10 mg	Myalgia, ↑ LFTs, ↑ glucose (small)	All 40–75 • **CI pregnancy**
Aspirin	75–162 mg/day (pregnancy 81–162 mg from 12–16 wk)	Bleeding, GI	Secondary prevention • preeclampsia prevention
Pregabalin	50 mg TID → 300–600 mg/day	Dizziness, sedation, **edema**, weight gain	Renal adjust • C-V • neuropathic pain
Duloxetine	30 → 60 mg daily	Nausea, dry mouth, hepatotoxicity	Avoid CrCl < 30 • **CYP1A2/2D6** (2D6 inhibitor) • serotonin syndrome
Amitriptyline	10–75 mg at bedtime	Anticholinergic, sedation, QT, orthostasis	Avoid elderly/CVD
Capsaicin 8% patch	Applied up to 1 h every 3 months	Burning, erythema	Topical option
Metoclopramide	5–10 mg before meals and bedtime	**Tardive dyskinesia (boxed)**, EPS	Gastroparesis • **≤ 12 weeks**

Drug	Dose	Key ADRs	Key notes / interactions
Sildenafil / tadalafil	50 mg / 10 mg as needed	Headache, flushing, hypotension	**CI with nitrates**

Exam Traps and Frequently Tested Points

1. **Diagnosis:** A1C ≥ 6.5% • FPG ≥ 126 • 2-h OGTT ≥ 200 • random ≥ 200 + symptoms • **prediabetes 5.7–6.4 / 100–125 / 140–199** • needs 2 abnormal tests.
2. **A1C unreliable:** anemia, hemoglobinopathies, pregnancy, transfusion, EPO, dialysis • **A1C 7% = eAG 154 mg/dL**.
3. **Screen all adults ≥ 35** (or overweight + risk factors) every 3 years • prior GDM → every 3 years for life.
4. **Goals:** A1C < 7% • preprandial 80–130 • postprandial < 180 • **BP < 130/80 • TIR > 70%**, TBR < 4%.
5. **DPP lifestyle: 7% weight loss + 150 min/week** → ↓ 58% • metformin for prevention: BMI ≥ 35, < 60 years, prior GDM.
6. **ASCVD → GLP-1 RA or SGLT2i with proven benefit • HF → SGLT2i • CKD → SGLT2i (eGFR ≥ 20)** + finerenone + GLP-1 RA • obesity → tirzepatide/semaglutide.
7. **A1C ≥ 1.5% above goal → dual therapy • A1C > 10% or BG ≥ 300 or catabolic → insulin** • GLP-1 RA before insulin.
8. **Metformin:** CI eGFR < 30 • don't start 30–45 • **B12 deficiency** • GI → titrate with food • hold for contrast (restart 48 h).
9. **SGLT2i:** genital mycotic infections, UTI, volume depletion, **euglycemic DKA**, Fournier's • **canagliflozin → amputation/fractures** • hold 3 days pre-op (ertugliflozin 4).
10. **GLP-1 RA: boxed MTC/MEN2** • GI effects • pancreatitis • gallbladder • **Rybelsus: empty stomach, ≤ 4 oz water, 30 min** • tirzepatide ↓ OC efficacy.
11. **Never combine DPP-4i with GLP-1 RA • linagliptin: no renal adjustment • saxagliptin/alogliptin → HF** • DPP-4i: arthralgia, bullous pemphigoid.
12. **TZDs:** boxed HF (CI NYHA III–IV), edema, weight gain, fractures, bladder cancer (pio), slow onset 2–3 months • gemfibrozil → pio max 15 mg.
13. **Sulfonylureas:** hypoglycemia + weight gain • **glyburide avoid in elderly/CKD → glipizide preferred** • CYP2C9 inhibitors (fluconazole, SMX/TMP) ↑ hypoglycemia.
14. **Repaglinide + gemfibrozil = CI** • glinides: skip meal, skip dose.
15. **Acarbose:** take with first bite • flatulence • **treat hypoglycemia with glucose, not sucrose**.
16. **Pramlintide:** ↓ prandial insulin by 50% • boxed severe hypoglycemia.
17. **Insulin kinetics:** rapid 15 min / 1–2 h / 3–5 h • regular 30 min / 2–3 h / 5–8 h • **NPH 1–2 h / 4–12 h / 12–18 h** • glargine flat ~24 h • degludec ~42 h.
18. **Clear before cloudy** (regular then NPH) • **never mix glargine/detemir/degludec • only regular (and rapid analogs) IV**.
19. **In-use insulin ~28 days** at room temperature (degludec/Toujeo 56) • unopened in fridge • rotate sites (lipohypertrophy).
20. **T1DM TDD 0.5 units/kg** (0.4–1) • **50% basal / 50% bolus • ICR 500/TDD • ISF 1800/TDD**.
21. **T2DM basal start 10 units or 0.1–0.2 units/kg** • ↑ 2 units every 3 days • hypoglycemia → ↓ 10–20% • **> 0.5 units/kg = overbasalization** → add GLP-1 or prandial.
22. **NPH BID → glargine = 80%** of total NPH • prandial start 4 units or 10% of basal at largest meal.
23. **Hypoglycemia:** level 1 < 70 • **level 2 < 54** • level 3 needs assistance • **Rule of 15** (4 glucose tabs / 4 oz juice) • glucagon 1 mg or nasal 3 mg.
24. **Beta-blockers mask hypoglycemia** except **sweating** • hypoglycemia unawareness → relax targets for weeks.
25. **DKA:** glucose ≥ 200, **pH < 7.3 / HCO3 < 18**, ketones ≥ 3, anion gap • **HHS:** glucose ≥ 600, **osm > 300**, no significant ketosis, older T2DM.
26. **DKA order: fluids first → check K (< 3.5 hold insulin) → regular insulin 0.1 units/kg/h → dextrose when BG < 250** (HHS < 300) → bicarb only pH < 7.0.
27. **DKA resolved:** ketones < 0.6, pH ≥ 7.3 or HCO3 ≥ 18 • **give SC basal 1–2 h before stopping the drip**.
28. **SGLT2i → euglycemic DKA** (glucose may be < 250) • children → cerebral edema.

29. **Statins:** all 40–75 (moderate) • high-intensity if risk factors or ASCVD • **ASCVD LDL < 55** • aspirin 75–162 mg for secondary prevention.
30. **UACR yearly** (T1 after 5 years, T2 at diagnosis) • **≥ 30 mg/g → ACEi/ARB** • SGLT2i eGFR ≥ 20 • finerenone (K) • **never ACEi + ARB**.
31. **Eye exam:** T1 within 5 years, T2 at diagnosis, then every 1–2 years • anti-VEGF • pregnancy → each trimester.
32. **Neuropathy:** monofilament yearly • **pregabalin, duloxetine**, TCAs, Na channel blockers • avoid opioids • check B12 (metformin).
33. **Gastroparesis:** metoclopramide ≤ 12 weeks (TD boxed) • stop GLP-1 RA/pramlintide • **ED: PDE5i — CI nitrates**.
34. **Feet:** daily self-inspection, no barefoot, no lotion between toes • comprehensive exam yearly.
35. **GDM:** screen 24–28 weeks • one-step **92 / 180 / 153** • two-step 50-g ≥ 130–140 → 100-g (95/180/155/140, ≥ 2 values).
36. **Pregnancy goals: fasting < 95**, **1-h < 140**, **2-h < 120**, A1C < 6% • **insulin preferred** • metformin and glyburide cross placenta • **CI ACEi/ARB/statins** • aspirin from 12–16 weeks.
37. **Postpartum GDM → 75-g OGTT at 4–12 weeks** • lifelong screening every 1–3 years • insulin needs fall sharply after delivery.
38. **Hospital:** target 140–180 • **basal-bolus**, **avoid sliding scale alone** (Beers) • ICU → IV insulin • steroid hyperglycemia → **NPH with the morning steroid**.
39. **Sick days:** never stop basal insulin (T1) • check ketones • hold **SADMANS** (SU, ACEi, diuretics, metformin, ARBs, NSAIDs, SGLT2i).
40. **Drugs ↑ glucose:** steroids, olanzapine/clozapine, thiazides, tacrolimus, protease inhibitors, niacin, statins, ICIs • **fluoroquinolones and pentamidine → both hypo and hyper**.
41. **Elderly:** avoid glyburide/glimepiride and sliding scale alone • A1C < 7–7.5% (healthy) / < 8% (complex) • deintensify.
42. **Vaccines:** influenza, pneumococcal, **hepatitis B (19–59)**, COVID, zoster ≥ 50, RSV (older adults).

> **Note:** doses and cut-offs follow the current ADA Standards of Care and the 2024 ADA/EASD hyperglycemic crises consensus. If your course slides or reference use slightly different numbers, **use your course numbers for the exam**.